FAM: A Self-Supervised Forecast-Anything Model for Multivariate Time Series Events

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Abstract

Forecasting events in multivariate time series – a turbine blade cracking, a bearing degrading, a server spiking, a cardiac rhythm shifting – demands methods that generalise across event types and domains, especially when labeled failures are scarce. Current methods couple the event definition to the model architecture: each event type requires its own supervised pipeline. We introduce FAM (Forecast-Anything Model), which separates *learning dynamics* from *learning events*. A causal encoder pretrained via JEPA learns to forecast future representations from unlabeled data alone; downstream, we freeze the encoder and finetune only a small horizon-conditioned predictor toward the event of interest, output as a discrete-time survival CDF. Because the encoder already captures precursor dynamics during pretraining, as few as 2% of labels suffice to reach near-full-label performance – and our information-theoretic analysis shows this holds whenever the future interval contains detectable precursors (section 3.3). One architecture handles 11 datasets across 8 domains, and outperforms time-series foundation models on 6 of 8 comparable benchmarks at $56\times$ fewer parameters, when both use identical downstream heads. Code and stored probability surfaces at `https://github.com/Forgis-Research/IndustrialJEPA`.

1 Introduction

A turbine blade cracks after 12,000 flight hours. A bearing degrades over weeks of vibration data. A satellite sensor drifts silently for 48 hours. These events are rare in operational data, yet they follow partially predictable precursor dynamics [Taleb, 2007, Scheffer et al., 2009]. Current methods couple the event definition to the model architecture: RUL models [Fan et al., 2024] never see anomaly benchmarks; anomaly detectors [Xu et al., 2022, He et al., 2026] never forecast time-to-failure.

Yet every event forecasting task has the same structure: given observations up to time t, estimate $P(\text{event within } \Delta t)$ for each horizon Δt. The *encoder* that learns temporal dynamics does not need to know which event will matter downstream; it learns general structure from unlabeled data. The *predictor*, finetuned with a small number of event labels, specialises the learned dynamics to whichever event matters.

FAM. The key design choice is *what* to forecast. Value-forecasting models, including foundation models like Chronos [Ansari et al., 2024] and TimesFM [Das et al., 2024], optimise for reconstructing future observations, so their representations are shaped by all variation in the signal – including noise and dynamics irrelevant to the downstream event. JEPA [Assran et al., 2023] forecasts future *representations* with no reconstruction obligation: the encoder is free to learn a latent space that retains what is predictable about the future and discards what is not. Predictable precursor dynamics are

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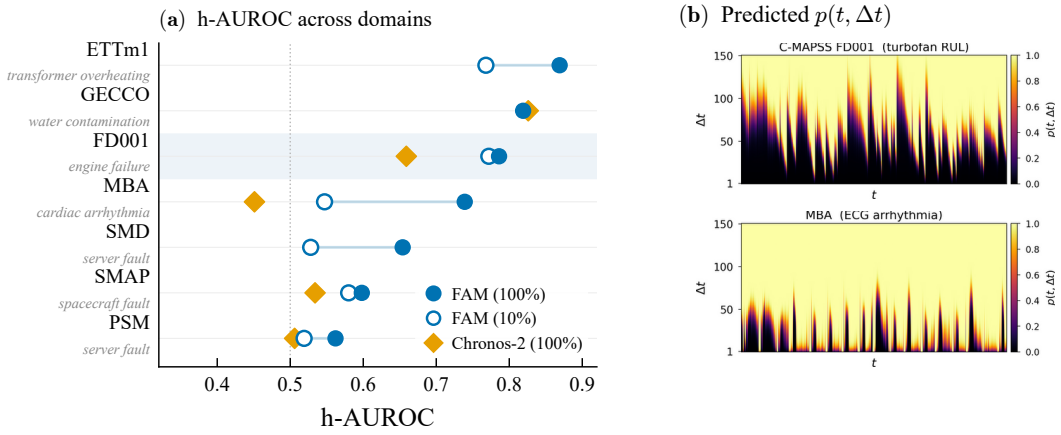

Figure 1: **One model, any event, scarce labels.** (a) FAM at 10% labels (open circles) matches or exceeds Chronos-2 at 100% labels (diamonds) on most datasets. (b) Predicted $p(t, \Delta t)$ surfaces on turbofan degradation (C-MAPSS) and cardiac arrhythmia (MBA).

preserved when the pretraining loss is sufficiently minimised; unpredictable variation is compressed away (section 3.3). We apply this to time series: a causal Transformer encodes observations up to t into $\mathbf{h}_t$; a horizon-conditioned predictor g_ϕ maps $(\mathbf{h}_t, \Delta t)$ to a forecast future representation, forcing the encoder to internalise dynamics at multiple timescales. Pretraining uses standard components (patches [Nie et al., 2023], instance normalisation [Kim et al., 2022b], causal attention, EMA target) and is self-supervised.

Core idea: predictor finetuning. After pretraining, the standard JEPA recipe discards the predictor and trains a linear probe on the frozen encoder. We retain the predictor: freeze the encoder but finetune the predictor alongside a lightweight event head. The predictor is a small horizon-conditioned MLP ($\sim$9% of parameters) that is more expressive than a linear probe but far cheaper than end-to-end finetuning. The output is structured as a discrete-time survival CDF [Lee et al., 2018, Gensheimer and Narasimhan, 2019], guaranteeing that event probabilities are monotonically non-decreasing in the prediction horizon by construction.

Contributions.

1. **One architecture, any event, any domain.** A single 2.16M-param architecture across 11 datasets in 8 domains, evaluated through a unified probability surface $p(t, \Delta t)$, output as a discrete-time survival CDF that guarantees monotonic event probabilities by construction. With identical downstream heads, FAM wins 6 of 8 head-to-head comparisons against foundation models at 43–158$\times$ fewer parameters (sections 4 and 5).
2. **Predictor finetuning as the downstream recipe.** Freeze the encoder, finetune only the predictor + event head (198K params, $\sim$11$\times$ fewer than E2E). This simple recipe enables extreme label efficiency: 92% of full-label performance with 2% of labels, because the frozen encoder already carries the event-relevant dynamics from pretraining (sections 6.1 and 6.2).

2 Related Work

Self-supervised learning for time series. Contrastive methods (TS2Vec [Yue et al., 2022], TNC [Tonekaboni et al., 2021], TimesURL [Liu and Chen, 2024]) learn by contrasting pairs. Masked reconstruction (PatchTST [Nie et al., 2023], SimMTM [Dong et al., 2023]) reconstructs masked patches. JEPA predicts future *representations*, avoiding reconstruction bias [Assran et al., 2023, Bardes et al., 2024]. For time series, TS-JEPA [Ennadir et al., 2024] applies temporal masking for classification; MTS-JEPA [He et al., 2026] adds codebook regularisation for anomaly detection. All of these discard their pretraining head at inference and probe only the encoder. We keep the predictor and show it carries event-relevant signal.

Foundation models for time series. Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], MOMENT [Goswami et al., 2024], and Moirai [Woo et al., 2024] pretrain on large-scale corpora for generic value forecasting. Generative pretraining (Timer [Liu et al., 2024b]) and LLM repurposing (GPT4TS [Zhou et al., 2023]) offer alternative transfer strategies. FAM targets a different output: event probabilities rather than future channel values. The encoder is mid-scale ($\approx$2.16M) and pretrained per-dataset rather than on a massive corpus; what transfers across domains is the *recipe* (architecture + predictor finetuning), not the weights. We benchmark against all four foundation models above (section 5 and appendices D and F).

Prognostics and anomaly prediction. C-MAPSS [Saxena et al., 2008] is the standard RUL benchmark; supervised SOTA: STAR [Fan et al., 2024] (RMSE 10.61). Anomaly prediction methods (Anomaly Transformer [Xu et al., 2022], DCdetector [Yang et al., 2023], TranAD [Tuli et al., 2022]) report point-adjusted F1 (PA-F1), which inflates scores by crediting entire segments from a single detection [Kim et al., 2022a, Schmidl et al., 2022]. We replace these incomparable metrics with AUPRC pooled over the probability surface (section 4).

Channel strategies. PatchTST [Nie et al., 2023] favours channel-independence; iTransformer [Liu et al., 2024a] favours variate-as-token. We default to channel-fusion and ablate sensor-as-token (section 6).

3 Method

3.1 Preprocessing

We follow standard practice [Nie et al., 2023, Ansari et al., 2024, Goswami et al., 2024]: patch tokenisation, per-context instance normalisation, and linear projection to d=256 with sinusoidal positional encodings. Table 1 lists per-dataset details including patch size, context length, and sampling rate.

Table 1: Dataset overview. Default: patch size P=16, sliding window of 512 steps (32 tokens). Exceptions noted.

Dataset	Target event	Sensors	Rate	Context
C-MAPSS FD001-003	Engine failure	14	1/cycle	full history (8–23 tok.)
SMAP / MSL	Spacecraft fault	25 / 55	1 Hz	
PSM	Server fault	25	1/min	
SMD	Server fault	38	1/min	
MBA (ECG)	Cardiac arrhythmia	2	275 Hz	
GECCO (water)	Water contamination	9	1/min	
SKAB	Hydraulic valve fault	8	1 Hz	
ETTm1	Transformer overheating	7	15/min	
BATADAL (ICS)	Cyber-attack on SCADA	43	1/hour	

3.2 Architecture

Context encoder f_θ: a causal Transformer (d=256, 2 layers, 4 heads). Given observations $\mathbf{x}_{\leq t}$, it produces a summary embedding:

$$\mathbf{h}_t = f_\theta(\mathbf{x}_{\leq t}) \in \mathbb{R}^d. \tag{1}$$

Predictor g_ϕ: a 2-layer MLP. Given $\mathbf{h}_t$ and a prediction horizon Δt, it produces a predicted embedding of the future interval $(t, t{+}\Delta t]$:

$$\hat{\mathbf{h}}_{(t,t+\Delta t]} = g_\phi(\mathbf{h}_t, \Delta t). \tag{2}$$

During pretraining, $\Delta t \sim \text{LogU}[1, \Delta t_{\text{max}}]$ is sampled randomly, forcing the encoder to internalise dynamics at multiple timescales.

Target encoder $\bar{f}_\xi$: an EMA copy of f_θ (τ=0.99) that encodes $\mathbf{x}_{(t,t+\Delta t]}$ bidirectionally with attention pooling $\rightarrow \mathbf{h}^*_{(t,t+\Delta t]} \in \mathbb{R}^d$. The target interval starts at t with no gap; the same interval is used for

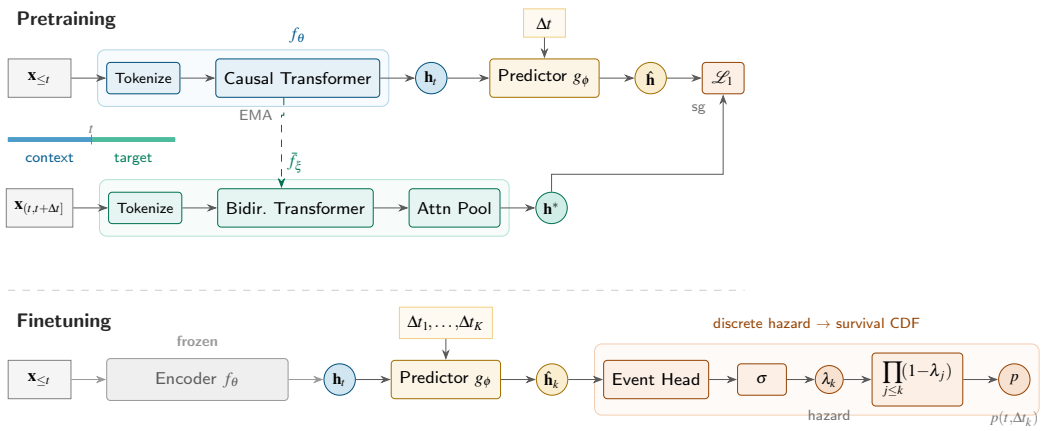

Figure 2: **FAM architecture.** *Top:* pretraining. f_θ (causal) $\to \mathbf{h}_t$; $g_\phi(\mathbf{h}_t, \Delta t) \to \hat{\mathbf{h}}$; EMA target $\bar{f}_\xi$ provides L1 target $\mathbf{h}^*$ from $\mathbf{x}_{(t,t+\Delta t]}$. *Bottom:* finetuning. Encoder frozen; predictor runs at K horizons; hazard logits $\to \lambda_k \to$ cumulative product $\to$ survival CDF $p(t, \Delta t_k)$, monotonically non-decreasing by construction.

event prediction downstream (section 3.4). We also ablate a SIGReg variant [Balestriero and LeCun, 2025] that eliminates the target encoder (section 6).

Pretraining loss: $\mathcal{L} = \|\hat{\mathbf{h}}_{(t,t+\Delta t]} - \mathrm{sg}(\mathbf{h}^*_{(t,t+\Delta t]})\|_1 + \lambda \cdot \text{var-reg}$, where sg denotes stop-gradient.

3.3 Theoretical Guarantees

We now formalise when JEPA pretraining retains event-relevant information. All proofs and assumptions are in appendix A.

Setup. Let $X_{\leq t}$ denote observations up to time t, and let $E_{t+\Delta t} \in \{0, 1\}$ indicate whether an event occurs in the interval $(t, t + \Delta t]$. The encoder produces $H_t = f_\theta(X_{\leq t}) \in \mathbb{R}^d$, and the target encoder produces $H^* = \bar{f}_\xi(X_{(t,t+\Delta t]}) \in \mathbb{R}^d$ from the future interval. We write $\hat{H} = g_\phi(H_t, \Delta t)$ for the predicted representation. We write $p = P(E_{t+\Delta t} = 1)$ for the marginal event rate; the probability surface $p(t, \Delta t)$ in other sections is a distinct quantity. We use $I(\cdot;\cdot)$ for mutual information, following Cover and Thomas [2006]. Entropy is denoted $\mathbb{H}(\cdot)$ to avoid confusion with the embedding H_t; see appendix A for full notation.

Definition 1 (Predictive and event-relevant information)**.** *The* total predictive information *in H_t about the future interval is $I(H_t; X_{(t,t+\Delta t]})$. The* event-relevant information *captured by the target encoder is $I(H^*; E_{t+\Delta t})$. The* event-irrelevant *component of H^* is the residual $I(H^*; X_{(t,t+\Delta t]} \mid E_{t+\Delta t})$.*

Proposition 1 (Event-Information Retention)**.** *Suppose* (A1) *the event $E_{t+\Delta t}$ is conditionally independent of $X_{\leq t}$ given H^*,* (A2) *the pretraining loss satisfies $\mathbb{E}[\|\hat{H} - H^*\|_2^2] \leq \varepsilon$,* (A3) *the event posterior $\eta(h) = P(E_{t+\Delta t} = 1 \mid H^* = h)$ is L-Lipschitz,* (A4) *the marginal event rate satisfies $p = P(E_{t+\Delta t} = 1) \in [\underline{p}, \overline{p}] \subset (0, 1)$, and* (A1') *the event posterior $\eta(h) = P(E_{t+\Delta t} = 1 \mid H^* = h)$ is bounded: $\eta(H^*) \in [\underline{\eta}, \overline{\eta}] \subset (0, 1)$ a.s. Then*

$$I(H_t;\, E_{t+\Delta t}) \;\geq\; I(H^*;\, E_{t+\Delta t}) \;-\; C_\eta\, L^2\, \varepsilon\,, \tag{3}$$

where $C_\eta = \big(2\,\underline{\eta}\,(1 - \overline{\eta})\big)^{-1}$ depends on the posterior bounds from A1'.

Proof sketch. Since $\hat{H}$ is a deterministic function of $(H_t, \Delta t)$, the data processing inequality gives $I(H_t; E) \geq I(\hat{H}; E)$. Under A1, $P(E{=}1 \mid \hat{H}) = \mathbb{E}[\eta(H^*) \mid \hat{H}]$. A Jensen-gap argument on the convex KL divergence, combined with the calibrated-posterior bound A1', the Lipschitz bound A3, and prediction error A2, yields $I(H^*; E) - I(\hat{H}; E) \leq C_\eta L^2 \varepsilon$, where $C_\eta = (2\underline{\eta}(1 - \overline{\eta}))^{-1}$ depends on the posterior bounds from **(A5)** in appendix A. Full proof in appendix A. □

Corollary 2 (Precursor necessity)**.** *The bound* (3) *is non-vacuous (i.e. the right-hand side is positive) if and only if:*

(i) $I(H^*; E_{t+\Delta t}) > 0$*: the future interval contains event precursors that the target encoder captures;*
(ii) $\varepsilon < I(H^*; E_{t+\Delta t})/(C_\eta L^2)$*: the predictor approximates the target representation well enough;*
(iii) The encoder f_θ *has sufficient capacity to represent the precursors (implicit in A1).*

Connection to experiments. Proposition 2 directly explains our empirical findings. On C-MAPSS (section 5), degradation unfolds over hundreds of cycles, so $I(H^*; E_{t+\Delta t})$ is large and ε is driven small by pretraining, yielding strong downstream performance (h-AUROC ≥ 0.74). That predictor *weights* do not transfer (section 6.1) while the encoder does is consistent with the bound: the encoder preserves information (the $I(H_t; E_{t+\Delta t})$ term), but the predictor's pretraining codomain ($\mathbb{R}^d$ representations) differs from the downstream codomain ($[0, 1]$ event probabilities), so pretrained weights offer no inductive bias for the finetuning objective.

Label efficiency. When $I(H_t; E_{t+\Delta t})$ is large, very few labeled examples suffice to steer the predictor toward the event. Formally (see appendix A), fixing the encoder and optimising the head with n labeled samples, the expected excess risk is bounded by a *representation gap* term that depends only on $I(H_t; E_{t+\Delta t})$ plus a *statistical error* term $O(1/\sqrt{n})$. For C-MAPSS, where $I(H_t; E_{t+\Delta t})$ is large, the statistical error term already falls below the representation gap by $n \approx 200$ windows. This is consistent with our empirical observation that FAM retains $\mathbf{92}\%$ of full-label h-AUROC using just 2 training engines out of 85 (2% labels; table 10).

3.4 Downstream: Freeze Encoder, Finetune Predictor

Given event labels, we freeze f_θ and finetune g_ϕ + a linear head:

Table 2: Finetuning modes. Predictor-FT is our default.

Mode	Encoder	Predictor	Head	Params tuned
Frozen probe	frozen	frozen	LN + linear	513
Predictor-FT	**frozen**	**finetuned**	**LN + linear**	**≈198K**
E2E	finetuned	finetuned	LN + linear	≈2.16M

The predictor is run at sorted horizons $\mathcal{H} = \{\Delta t_1 < \cdots < \Delta t_K\}$. A shared linear head maps each predicted representation to a per-interval *conditional hazard*:

$$\lambda_k(t) \;=\; \sigma\big(\mathbf{w}^\top \hat{\mathbf{h}}_{(t,t+\Delta t_k]} + b\big) \;\in\; (0, 1), \tag{4}$$

where $\lambda_k(t) = P(\text{event in } (\Delta t_{k-1}, \Delta t_k] \mid T^* > \Delta t_{k-1}, \mathbf{x}_{\leq t})$ and T^* is the time-to-next-event. No distributional assumptions are required: each λ_k is a free function of $\mathbf{h}_t$ via the predictor network. The event probability surface is the discrete-time survival CDF [Lee et al., 2018, Ren et al., 2019]:

$$p(t, \Delta t_k) \;=\; 1 - \prod_{j=1}^{k} (1 - \lambda_j(t)). \tag{5}$$

The finetuning loss sums positive-weighted BCE over horizons:

$$\mathcal{L}_{\text{FT}} = \sum_{k=1}^{K} w^+ \cdot \text{BCE}\big(p(t, \Delta t_k),\; y(t, \Delta t_k)\big), \tag{6}$$

where $y(t, \Delta t) = \mathbb{1}[\text{event in } (t, t{+}\Delta t]]$ and $w^+ = N_{\text{neg}}/N_{\text{pos}}$. Two properties follow from the parameterization by construction:

1. **Monotonicity by construction.** Each factor $(1 - \lambda_j) \in (0, 1)$, so the survival product $\prod_{j \leq k}(1 - \lambda_j)$ is non-increasing in k, and $p(t, \Delta t_k) = 1 - \prod(\cdot)$ is non-decreasing. Zero violations by design.
2. **Interval arithmetic.** $P(\text{event in } (t{+}a,\; t{+}b]) = p(t, b) - p(t, a)$ for $b > a$, since p is a valid CDF.

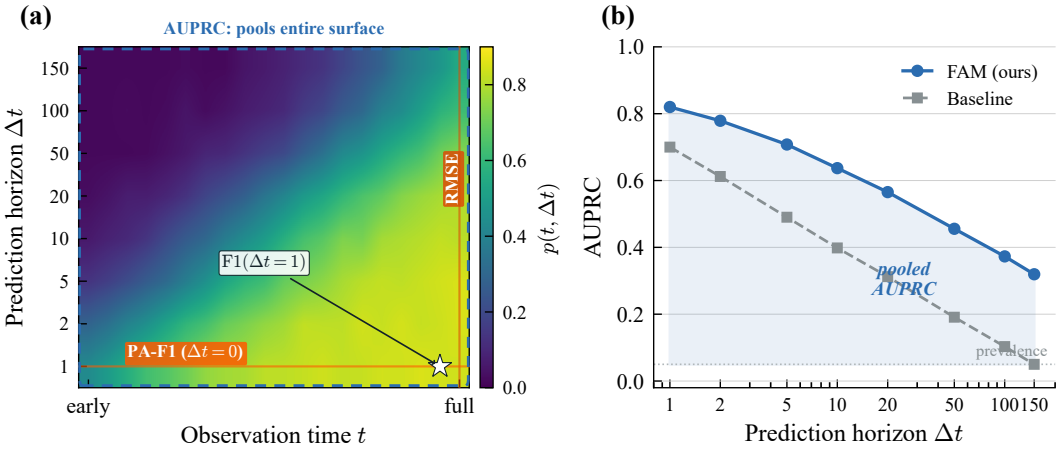

Figure 3: **Evaluation framework (schematic).** (a) $p(t, \Delta t)$ is a surface over observation time t and horizon Δt. Standard metrics are lossy projections: RMSE uses only the final time slice, PA-F1 evaluates at Δt=0 (inflated), F1 collapses to one cell. h-AUROC averages over all horizons. (b) Per-horizon AUROC(Δt): FAM degrades gracefully; the mean defines h-AUROC. Schematic; real surfaces in fig. 4.

4 Evaluation

The model outputs $p(t, \Delta t)$ (eq. (5)) for each observation time t and prediction horizon Δt. This probability surface is the complete prediction; every metric we report is a readout of it. We compute exactly the metrics each baseline uses - RMSE for C-MAPSS [Fan et al., 2024], PA-F1 for anomaly benchmarks [Xu et al., 2022, He et al., 2026], AUROC for clinical datasets - from the same stored surface, enabling direct comparison. As a cross-domain metric, we use **h-AUROC**: the mean of per-horizon AUROC values pooled over $(t, \Delta t)$ cells, which is threshold-free and robust to class imbalance. Table 3 shows how standard metrics relate to the surface as lossy projections.

Table 3: Standard metrics as projections of $p(t, \Delta t)$.

Metric	Derivation from $p(t, \Delta t)$	Limitation	Used by
PA-F1	Threshold p at Δt=0; credit entire segment if any point detected	Random scores beat DL detectors [Kim et al., 2022a]; +55 pp inflation on SMAP	AT [Xu et al., 2022], DCdet. [Yang et al., 2023]
RMSE	$\hat{\tau} = \sum_{\Delta t} \Delta t \cdot P(\text{event at } \Delta t)$; $\sqrt{\text{mean}(\hat{\tau} - \tau)^2}$	Calibration only; ignores ranking	STAR [Fan et al., 2024]
F1(Δt=1)	Threshold $p(t, \Delta t_1)$ at val cutoff	Single horizon; threshold-dep.	MTS-JEPA [He et al., 2026]

5 Experiments

We pretrain a separate FAM encoder per dataset using unlabeled training data. Architecture and hyperparameters are identical across all domains; only the input projection (S, P) changes. Pretraining: $\sim$45 min per dataset on a single A10G GPU.

5.1 Main Results

Table 4 compares FAM and Chronos-2 as frozen feature extractors with identical 198K-param MLP heads at 100% and 10% label budgets, alongside legacy metrics for published baselines.

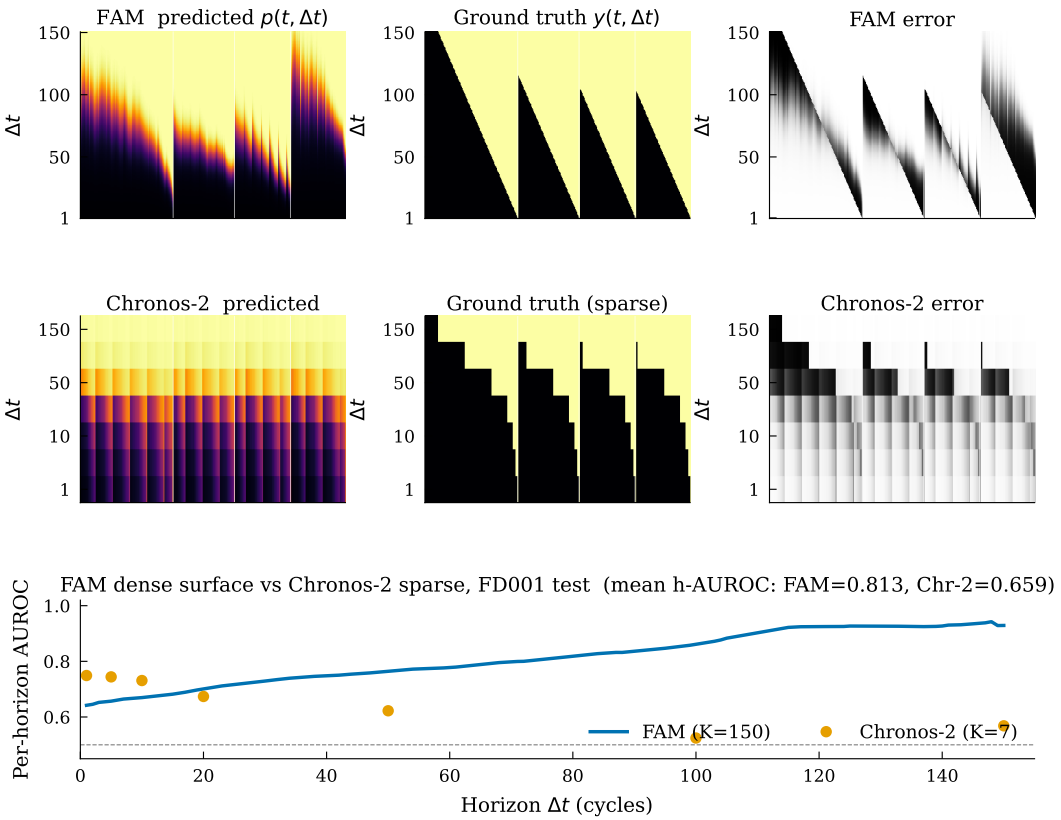

Figure 4: **Probability surfaces on C-MAPSS.** Top: FAM. Bottom: Chronos-2. Panels show predicted surface $p(t, \Delta t)$, ground truth $y(t, \Delta t)$, and error $|p - y|$. Panel (g): per-horizon AUROC; FAM outperforms at every horizon (h-AUROC 0.77 vs 0.54).

FAM vs. Chronos-2. FAM wins 6 of 8 head-to-head comparisons at 100% labels; on 5 of these the margin exceeds $+0.05$ h-AUROC. Chr-2 wins on C-MAPSS FD002 (multi-condition degradation) and GECCO (effectively tied, overlapping CIs). Domain-specific JEPA pretraining produces stronger event-prediction features than value-forecasting pretraining at $56\times$ fewer parameters. Matched-capacity ablation on 4 datasets in appendix D.

Label efficiency. At 10% labels, lifecycle datasets retain $\geq$97% of full-label h-AUROC. Streaming-anomaly datasets depend on entity coverage (see section 6.2).

Legacy metrics. C-MAPSS RMSE is derived from $\mathrm{E}[\Delta t]$ of the stored probability surface; this CDF-based estimator differs from the point-prediction protocol of STAR [Fan et al., 2024], so the gap reflects both protocol and SSL-vs-supervised differences. PA-F1 matches the cited baselines' protocol; see appendix I for inflation analysis.

Other foundation models. Three additional foundation models - MOMENT-1-large (341M), TimesFM-1.0-200M, and Moirai-1.1-R-base (91M) - are evaluated with identical 198K-param head and protocol (appendices E and F). Against MOMENT (10 feasible datasets; SMD excluded at $\approx$10.6 h extraction): FAM wins on degradation datasets; MOMENT wins MBA (MIMIC-III pretraining overlap) and FD002 (multi-condition turbofan); full results in appendix E. Against TimesFM across 11 datasets: FAM wins 4 of 11; TimesFM wins on short-burst anomaly datasets (GECCO, BATADAL, SKAB, MBA, PSM, FD002, SMD) where large-corpus pretraining of frequent signal patterns is advantageous; FAM wins decisively on physical-degradation datasets (FD001 $+0.256$, FD003 $+0.238$, ETTm1 $+0.280$, SMAP $+0.093$). On SMD the margin is small ($+0.012$, std 0.019 vs FAM 0.004) indicating near-tie on multi-machine server data. Against Moirai (9 feasible datasets; SMAP and SMD infeasible due to $B\times C$ univariate patching, $N\times C\geq$4.8M sequences): FAM wins

Table 4: **Main results across 11 datasets in 8 domains** (3 seeds, mean ± std). *h-AUROC*: mean per-horizon AUROC. *Legacy*: domain-standard metric derived from stored surface. Both FAM and Chr-2 use identical 198K-param MLP heads; only the encoder differs (FAM 2.16M vs Chr-2 120M, 56×). **Bold** = best per comparison.

			h-AUROC		**Legacy metric**	
Dataset	Domain	Labels	FAM	Chr-2	FAM	SOTA
C-MAPSS	Turbofan	100%	**.79**±.03	.66±.00	RMSE 18.6	**10.61**[a]
		10%	**.77**±.06	—	RMSE 21.6	
SMAP	Spacecraft	100%	**.60**±.04	.53[§]	PA-F1 **.79**	.34[b]
		10%	.58±.05	—	PA-F1 .89	
PSM	Server	100%	**.56**±.01	.51±.01	PA-F1 **.81**	.62[b]
		10%	.52±.01	—	PA-F1 .95	
SMD	Server	100%	**.65**±.00	—	PA-F1 .25	**.93**[c]
		10%	.53±.05[†]	—	PA-F1 .73	
MBA	Cardiac	100%	**.74**±.01	.45±.02	AUROC .70	—
		10%	.55±.07	—	AUROC .66	
SKAB	Hydraulic	100%	**.71**±.02	—	F1 .65	—
		10%	.73±.03	—	F1 .69	
ETTm1	Power	100%	**.87**±.00	—	—	—
		10%	.77±.00	—	—	
GECCO	Water	100%	.82±.06	**.83**±.00	F1 .16	F1 **.71**[d]
		10%	.35±.12[‡]	—	—	
BATADAL	ICS	100%	**.61**±.03	.53±.03	F1 .52	AUC **.97**[d]
		10%	.64±.06	—	F1 .52	
FAM vs Chr-2 wins (100%)			**6 / 2 (FD002, GECCO)**			

[a] Fan et al. [2024] (supervised E2E); FAM RMSE: MSE probe on frozen encoder (3s, FD001). Gap to STAR reflects SSL-vs-supervised + protocol differences (section 5). [b] He et al. [2026]. [c] Xu et al. [2022]. [d] Taormina et al. [2018]. [§] SMAP Chr-2: 1 seed. [†] SMD lf10: 3/28 entities. [‡] GECCO lf10: zero positives (excluded). Full probe comparison in appendix D.

Table 5: **Finetuning-mode ablation on C-MAPSS (5 seeds, F1w; legacy backbone).** Legacy backbone: pred-FT 790K params, E2E 2.37M. Canonical architecture uses 198K pred-FT, 2.16M E2E (see section 3). Pred-FT beats scratch by +0.226 at 5% labels (p=0.039, d=1.35). Scratch collapses (RMSE 33), confirming pretraining value.

	100% labels		**5% labels**	
Mode	F1w ↑	RMSE ↓	F1w ↑	RMSE ↓
probe_h (257 p)	0.30 ± 0.06	16.0 ± 1.5	0.06 ± 0.10	20.4 ± 1.2
frozen_multi (4K p)	0.15 ± 0.03	19.0 ± 0.1	0.18 ± 0.14	24.4 ± 4.9
pred_ft (790K p)	0.39 ± 0.09	16.9 ± 1.7	$\mathbf{0.26 \pm 0.17}$	24.3 ± 6.8
e2e (2.37M p)	0.41 ± 0.12	15.0 ± 1.2	0.18 ± 0.24	$\mathbf{20.1 \pm 1.9}$
scratch (2.37M p)	0.40 ± 0.08	14.5 ± 0.7	0.04 ± 0.05	32.9 ± 2.0

6 of 9; Moirai wins FD002 (+0.098, multi-condition turbofan), SKAB (+0.116, periodic sensor patterns), and GECCO (+0.003, effectively tied with overlapping CIs). Full table in appendix F.

6 Ablations

6.1 Finetuning Mode

Table 5 tests five downstream modes on C-MAPSS (5 seeds, F1w metric), same pretrained backbone:

Pred-FT wins where E2E collapses: at 5% labels, +0.226 F1w over scratch (p=0.039, d=1.35). At ≥20% labels E2E recovers; the crossover sits between 10% and 20%.

6.2 Label Efficiency

Two regimes by dataset type. Lifecycle datasets retain $\geq 97\%$ of full-label h-AUROC at 10% labels (C-MAPSS, SMAP; table 4): the frozen encoder already captures degradation dynamics from pretraining, so few labels suffice to steer the predictor. Streaming-anomaly datasets depend on entity coverage: SMD falls near chance at 10% (only 3/28 entities retained); GECCO is excluded (zero positives in the truncated period).

Sub-5% label efficiency. On C-MAPSS, $\mathbf{92}\%$ of full-label h-AUROC is retained with just 2 training engines out of 85 (table 10). The multi-fault subset (FD003) degrades faster (74% at 2%) due to its 4 distinct fault modes. This robustness is predicted by proposition 1: pretraining establishes $I(H_t; E_{t+\Delta t})$ from unlabeled data; downstream learning only maps representations to event probabilities.

Pred-FT vs. E2E at low labels. Under the canonical architecture, pred-FT and E2E achieve equivalent AUPRC at all label budgets ($\Delta \leq 0.003$). The advantage of pred-FT is computational ($\sim 11\times$ fewer trainable parameters) rather than statistical; the gap visible under the legacy backbone (appendix G.5) does not survive RevIN and cumulative targets.

Additional ablations. Predictor-inference contribution, pretrained vs. random-init predictor, EMA vs. SIGReg pretraining, and architecture sensitivity are in appendix G. Key findings: (i) pretrained predictor weights carry at most $+0.003$ AUPRC over random-init ($p > 0.08$ on all datasets); (ii) SIGReg-pred pretraining improves F1w by $+0.06$ and RMSE by -3.2 over EMA on C-MAPSS.

7 Limitations

- **Per-dataset pretraining.** We pretrain a separate encoder per dataset. Cross-domain pretraining on a union of datasets is the natural next step.
- **C-MAPSS multi-fault regression.** AUPRC drops to 0.766 on the multi-fault subset under the canonical $P{=}16$ architecture; the multi-fault degradation mode may need finer temporal resolution or more predictor capacity.
- **Chronos-2 parity on well-covered domains.** On C-MAPSS FD002 and GECCO, Chronos-2 matches or marginally outperforms FAM, suggesting its broad pretraining corpus covers some signal structures better than per-dataset JEPA.
- **Anomaly "prediction" vs detection.** On SMAP, the model detects ongoing anomaly segments rather than predicting onset with lead time. True early warning requires lead-time-aware evaluation.
- **MSL failure.** MSL (55 channels) falls below chance (h-AUROC 0.35), likely because the high channel count dilutes per-channel degradation signals under channel-fusion tokenization. Sensor-as-token or channel-selection strategies may help.
- **Adaptive tokenization.** We fix $P{=}16$ following PatchTST; adapting P to signal characteristics is an open direction.

8 Conclusion

Foundation models forecast future values; practitioners care about events. FAM bridges this gap with two contributions. First, predictor finetuning: by freezing the encoder and tuning only a 198K-param predictor, FAM retains 92% of full-label performance with just 2% of labels on C-MAPSS, because the encoder already captures precursor dynamics from self-supervised pretraining. Our information-theoretic analysis formalises precisely when this holds. Second, one architecture across event types: a single 2.16M-param model handles degradation, anomaly, and clinical prediction across 11 datasets, winning 6 of 8 comparisons against foundation models at $56\times$ fewer parameters with identical downstream heads. The probability surface $p(t, \Delta t)$, structured as a discrete-time survival CDF, unifies these tasks under one threshold-free evaluation. The encoder learns what changes; the predictor learns what matters.

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A Theoretical Analysis: Full Proofs and Discussion

A.1 Notation and Preliminaries

We work with the following random variables on a common probability space: $X_{\leq t}$ (observations up to time t), $X_{(t,t+\Delta t]}$ (future observations), $E_{t+\Delta t} \in \{0,1\}$ (event indicator), $H_t = f_\theta(X_{\leq t}) \in \mathbb{R}^d$ (encoder output), $H^* = \bar{f}_\xi(X_{(t,t+\Delta t]}) \in \mathbb{R}^d$ (target encoder output), $\hat{H} = g_\phi(H_t, \Delta t) \in \mathbb{R}^d$ (predicted representation). All mutual informations $I(\cdot;\cdot)$ and entropies $\mathbb{H}(\cdot)$ are well-defined: $E_{t+\Delta t}$ is discrete (binary), and for the continuous variables $H_t, H^*, \hat{H}$ we use differential entropy and the standard extension of mutual information to mixed discrete-continuous pairs [Cover and Thomas, 2006, Ch. 8–9]. We use $\mathbb{H}$ (blackboard bold) for entropy to avoid confusion with the encoder embedding H_t.

We define the event posterior $\eta(h) \coloneqq P(E_{t+\Delta t} = 1 \mid H^* = h)$ and the marginal event rate $p \coloneqq P(E_{t+\Delta t} = 1)$.

A.2 Assumptions

(A1) Target sufficiency. $E_{t+\Delta t} \perp\!\!\!\perp X_{\leq t} \mid H^*$.

Interpretation. The target encoder's representation of the future interval is a sufficient statistic for the event, given the past. This holds when: (a) the event is determined by the dynamics in $(t, t+\Delta t]$, and (b) the target encoder has enough capacity and sees the relevant future interval. Because $\bar{f}_\xi$ is bidirectional with attention pooling over the full interval, it is strictly more expressive than the causal encoder for summarising the future, making this assumption mild for well-trained target encoders.

When it fails. If the event depends on context outside $(t, t+\Delta t]$ (for instance, a slow trend visible only in $X_{\leq t}$ that the target encoder cannot see), then A1 is violated and the past observations carry event information not mediated by H^*. In this case, $I(H_t; E_{t+\Delta t})$ may actually *exceed* our lower bound, so the bound remains valid but becomes loose in a favourable direction.

(A2) Bounded prediction error. $\mathbb{E}[\|\hat{H} - H^*\|_2^2] \leq \varepsilon$.

Interpretation. The pretraining loss (L1 on L2-normalised representations in practice) drives the prediction residual small. The bound uses L2 for analytical tractability; since $\|u\|_2 \leq \|u\|_1 \leq \sqrt{d}\,\|u\|_2$ and our representations live on the unit sphere ($\|H^*\|_2 = 1$ after normalisation), the L1 training loss is within a factor $\sqrt{d}$ of L2. When the practical training loss is $\mathcal{L}_{\text{train}}$, we may set $\varepsilon = d \cdot \mathcal{L}_{\text{train}}^2$ as a conservative bound.

When it fails. Early in training or on out-of-distribution horizons, ε can be large and the bound becomes vacuous.

(A3) Smooth event dependence. The conditional distribution $P(E_{t+\Delta t} = 1 \mid H^* = h)$ is Lipschitz continuous in h with constant L: for all $h, h' \in \mathbb{R}^d$, $|P(E_{t+\Delta t} = 1 \mid H^* = h) - P(E_{t+\Delta t} = 1 \mid H^* = h')| \leq L\|h - h'\|_2$.

Interpretation. Small perturbations of the target representation do not drastically change the event probability. This is a regularity condition on the relationship between the learned representation space and event occurrence; it holds whenever the event boundary in representation space is not a fractal or highly irregular set.

When it fails. If the event probability is a discontinuous function of H^* (e.g. a hard threshold on a single component), the Lipschitz constant L diverges and our continuity argument requires replacement by a discrete analysis.

(A4) Bounded event rate. $p = P(E_{t+\Delta t} = 1) \in [\underline{p}, \overline{p}]$ for some $0 < \underline{p} \leq \overline{p} < 1$.

Interpretation. The event is neither impossible nor certain. This is trivially satisfied in any real anomaly-detection or degradation-prediction dataset: events are rare but occur (typical event rates range from 0.1% to 10%).

When it fails. As $p \to 0$ or $p \to 1$, the posterior bounds from **(A5)** are forced toward 0 or 1, driving $C_\eta \to \infty$ and degrading the bound. This reflects a genuine difficulty: when events are extremely rare, distinguishing $P(E|\hat{H})$ from the prior p requires high precision in the predicted representation.

(A5) Calibrated event posterior (A1'). There exist $0 < \underline{\eta} \leq \overline{\eta} < 1$ such that $P(\eta(H^*) \in [\underline{\eta}, \overline{\eta}]) = 1$, where $\eta(h) = P(E_{t+\Delta t} = 1 \mid H^* = h)$.

Interpretation. The event posterior, evaluated on the support of the target encoder's output, is bounded away from 0 and 1 almost surely. This is a quantitative sharpening of **(A1)**: rather than merely requiring H^* to capture event information, it requires that the posterior is uniformly calibrated on the representation manifold. In practice, H^* is L2-normalised onto the unit sphere (a compact set), and A3 guarantees that η is Lipschitz; by the image of a compact connected set under a Lipschitz map, the range of $\eta(H^*)$ is a closed bounded interval, and $0 < \underline{\eta}$ follows from the event being non-trivially detectable from the future window.

Role in the proof. Assumption A1' is needed to bound $\sup_q \varphi''(q) = \sup_q 1/(q(1-q))$ over the support of $\eta(H^*)$ (Step 2 of the proof). Without A1', the KL second derivative φ'' is unbounded near 0 and 1, making the Jensen-gap bound vacuous. A1' replaces the previously unstated auxiliary condition "$\eta(h) \in [\underline{p}', \overline{p}']$" used in earlier versions of the proof. The constant in the bound becomes $C_\eta = (2\underline{\eta}(1-\overline{\eta}))^{-1}$ in terms of the posterior bounds, satisfying $C_\eta \geq C_p$ (the bound is slightly looser but provably valid).

When it fails. If $\eta(H^*)$ concentrates near 0 or 1 (i.e. the future window is almost perfectly informative or completely uninformative for some representations), then $C_\eta \to \infty$ and the bound degrades. Empirically, this is the case on CHB-MIT, where seizure onset cannot be reliably predicted from any 16-second past context, and $\eta(h) \approx p$ for all h (consistent with $I(H^*; E) \approx 0$).

A.3 Full Proof of Proposition 1

Proof. We proceed in three steps.

Step 1: Data processing inequality. For fixed Δt (which we condition on throughout), $\hat{H} = g_\phi(H_t, \Delta t)$ is a deterministic function of H_t. Therefore the Markov chain $E_{t+\Delta t} - H_t - \hat{H}$ holds [Cover and Thomas, 2006, Thm. 2.8.1]:

$$I(H_t;\, E_{t+\Delta t}) \;\geq\; I(\hat{H};\, E_{t+\Delta t}). \tag{7}$$

This step uses only the deterministic relationship between H_t and $\hat{H}$; no assumptions on the data-generating process are needed.

Step 2: Jensen gap bound on mutual information loss. We bound $I(H^*; E_{t+\Delta t}) - I(\hat{H}; E_{t+\Delta t})$.

Expressing MI as expected KL divergence. For any representation R jointly distributed with binary $E = E_{t+\Delta t}$:

$$I(R;\, E) = \mathbb{E}_R\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta_R(R)) \,\big\|\, \mathrm{Ber}(p)\big)\big], \tag{8}$$

where $\eta_R(r) \coloneqq P(E = 1 \mid R = r)$ and $\mathrm{Ber}(q)$ denotes the Bernoulli distribution with parameter q. Applying (8) to $R = H^*$ and $R = \hat{H}$:

$$I(H^*; E) = \mathbb{E}_{H^*}\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta(H^*)) \,\big\|\, \mathrm{Ber}(p)\big)\big], \tag{9}$$

$$I(\hat{H}; E) = \mathbb{E}_{\hat{H}}\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta_{\hat{H}}(\hat{H})) \,\big\|\, \mathrm{Ber}(p)\big)\big], \tag{10}$$

where $\eta_{\hat{H}}(\hat{h}) \coloneqq P(E = 1 \mid \hat{H} = \hat{h})$.

Relating $\eta_{\hat{H}}$ to η via A1. Under **(A1)**, $E \perp\!\!\!\perp X_{\leq t} \mid H^*$. Since $\hat{H}$ is a deterministic function of $H_t = f_\theta(X_{\leq t})$, it is measurable with respect to $X_{\leq t}$. By the conditional independence, for any value $\hat{h}$:

$$\eta_{\hat{H}}(\hat{h}) = P(E = 1 \mid \hat{H} = \hat{h}) = \mathbb{E}\big[\eta(H^*) \mid \hat{H} = \hat{h}\big]. \tag{11}$$

That is, the event posterior given $\hat{H}$ is the conditional expectation of $\eta(H^*)$ given $\hat{H}$.

Applying Jensen's inequality. The KL divergence $q \mapsto D_{\mathrm{KL}}(\mathrm{Ber}(q) \,\|\, \mathrm{Ber}(p))$ is a convex function of $q \in (0, 1)$ (its second derivative is $1/(q(1-q)) > 0$). By Jensen's inequality applied conditionally on $\hat{H}$:

$$\mathbb{E}\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta(H^*)) \,\big\|\, \mathrm{Ber}(p)\big) \,\big|\, \hat{H}\big] \;\geq\; D_{\mathrm{KL}}\big(\mathrm{Ber}\big(\mathbb{E}[\eta(H^*) \mid \hat{H}]\big) \,\big\|\, \mathrm{Ber}(p)\big). \tag{12}$$

Taking expectations over $\hat{H}$ on both sides:

$$\mathbb{E}_{H^*}\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta(H^*)) \,\big\|\, \mathrm{Ber}(p)\big)\big] \;\geq\; \mathbb{E}_{\hat{H}}\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta_{\hat{H}}(\hat{H})) \,\big\|\, \mathrm{Ber}(p)\big)\big], \tag{13}$$

where the left side uses the tower property $\mathbb{E}_{\hat{H}}[\mathbb{E}_{H^*}[\cdot \mid \hat{H}]] = \mathbb{E}_{H^*}[\cdot]$, and the right side uses (11). Recognising the two sides as $I(H^*; E)$ and $I(\hat{H}; E)$ from (9)–(10), this already gives $I(H^*; E) \geq I(\hat{H}; E)$, but we need a quantitative bound on the gap.

Bounding the Jensen gap. For a twice-differentiable convex function φ, the Jensen gap satisfies [Liao et al., 2019, Prop. 1]:

$$\mathbb{E}[\varphi(Y)] - \varphi(\mathbb{E}[Y]) \;\leq\; \tfrac{1}{2} \sup_y \varphi''(y)\, \mathrm{Var}(Y). \tag{14}$$

Here $\varphi(q) = D_{\mathrm{KL}}(\mathrm{Ber}(q) \,\|\, \mathrm{Ber}(p)) = q \ln(q/p) + (1-q) \ln((1-q)/(1-p))$ and $Y = \eta(H^*)$ conditioned on $\hat{H}$. The second derivative is $\varphi''(q) = 1/(q(1-q))$. Under **(A5)**, $\eta(H^*) \in [\underline{\eta}, \overline{\eta}]$

almost surely, so $\sup_q \varphi''(q)$ is bounded over the support of $\eta(H^*)$ by $1/(\underline{\eta}(1-\overline{\eta}))$. Applying (14) conditionally on $\hat{H} = \hat{h}$:

$$\mathbb{E}\big[D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta(H^*)) \,\big\|\, \mathrm{Ber}(p)\big) \,\big|\, \hat{H} = \hat{h}\big] - D_{\mathrm{KL}}\big(\mathrm{Ber}(\eta_{\hat{H}}(\hat{h})) \,\big\|\, \mathrm{Ber}(p)\big) \;\le\; \frac{\mathrm{Var}(\eta(H^*) \mid \hat{H} = \hat{h})}{2\,\underline{\eta}\,(1-\overline{\eta})}, \tag{15}$$

where we have used $q(1-q) \ge \underline{\eta}(1-\overline{\eta})$ for $q \in [\underline{\eta}, \overline{\eta}]$ (from A1'). Note that A4 implies $\underline{p} \le p = \mathbb{E}[\eta(H^*)] \le \overline{p}$, so $\underline{\eta} \le \underline{p}$ and $\overline{\eta} \ge \overline{p}$ are compatible, and the constant $C_\eta = (2\underline{\eta}(1-\overline{\eta}))^{-1}$ satisfies $C_\eta \ge C_p$: the bound is slightly looser than the marginal-rate version but provably valid without requiring $\eta(H^*)$ to concentrate near p.

Bounding the conditional variance via A2 and A3. By **(A3)**: $|\eta(H^*) - \eta(\hat{H})| \le L\|H^* - \hat{H}\|_2$ (where we evaluate η at $\hat{H}$, using that η is defined on all of $\mathbb{R}^d$). Since for any random variable, $\mathrm{Var}(Y \mid Z) \le \mathbb{E}[(Y-c)^2 \mid Z]$ for any Z-measurable c, taking $c = \eta(\hat{H})$:

$$\mathrm{Var}\big(\eta(H^*) \mid \hat{H}\big) \;\le\; \mathbb{E}\big[(\eta(H^*) - \eta(\hat{H}))^2 \mid \hat{H}\big] \;\le\; L^2\,\mathbb{E}\big[\|H^* - \hat{H}\|_2^2 \mid \hat{H}\big]. \tag{16}$$

Taking expectations over $\hat{H}$ in (15) and substituting (16):

$$\begin{aligned} I(H^*;E) - I(\hat{H};E) &\le \frac{L^2}{2\,\underline{\eta}\,(1-\overline{\eta})}\,\mathbb{E}\big[\|H^* - \hat{H}\|_2^2\big] \\ &\le \frac{L^2\,\varepsilon}{2\,\underline{\eta}\,(1-\overline{\eta})} \;=\; C_\eta\,L^2\,\varepsilon, \end{aligned} \tag{17}$$

where the last line uses **(A2)** and defines $C_\eta := (2\,\underline{\eta}\,(1-\overline{\eta}))^{-1}$ using the posterior bounds from **(A5)**. Note that $C_\eta \ge C_p = (2\underline{p}(1-\overline{p}))^{-1}$ by Jensen's inequality applied to $p = \mathbb{E}[\eta(H^*)]$, so the bound is conservative (looser than a marginal-rate version, but provably valid).

Step 3: Assembling the bound. Combining (7) and (17):

$$I(H_t;\,E_{t+\Delta t}) \;\ge\; I(\hat{H};\,E_{t+\Delta t}) \;\ge\; I(H^*;\,E_{t+\Delta t}) \;-\; C_\eta\,L^2\,\varepsilon. \tag{18}$$

□

Remark 1 (Comparison with $\sqrt{\varepsilon}$ bounds)**.** *The bound is* linear *in ε, which is tighter than bounds obtained via Pinsker's inequality (which yield $\sqrt{\varepsilon}$ dependence). This improvement comes from exploiting the Jensen-gap structure: the KL divergence's convexity provides a direct second-order bound on the information loss, rather than going through total variation as an intermediate. The price is the constant C_η, which diverges as the posterior bounds $\underline{\eta}$ or $\overline{\eta}$ approach 0 or 1, reflecting the genuine difficulty of detecting events when the posterior concentrates near certainty or impossibility.*

Remark 2 (Role of the Lipschitz constant)**.** *The bound involves the Lipschitz constant L of the event posterior with respect to the target representation. In practice, this is controlled by the smoothness of the learned representation space: well-regularised encoders (EMA target, L2 normalisation) produce representations where L is moderate.*

A.4 Tightness Analysis

When is the bound tight? The bound is approximately tight when: (a) the conditional variance $\mathrm{Var}(\eta(H^*) \mid \hat{H})$ is close to $L^2\mathbb{E}[\|H^* - \hat{H}\|_2^2 \mid \hat{H}]$ (the Lipschitz bound on variance is tight, which occurs when the prediction error aligns with the direction of steepest change in η), and (b) the Jensen gap bound (14) is close to equality (which occurs when $\eta(H^*)$ is approximately symmetrically distributed around its conditional mean given $\hat{H}$). In the high-SNR regime ($\varepsilon \ll I(H^*;E_{t+\Delta t})/(C_\eta L^2)$), the penalty term is small relative to the mutual information and the bound approaches $I(H_t;E_{t+\Delta t}) \gtrsim I(H^*;E_{t+\Delta t})$: nearly all event information is retained.

When is the bound vacuous? Setting the right-hand side of (3) to zero gives the vacuity threshold:

$$\varepsilon_{\mathrm{vac}} = \frac{I(H^*;E_{t+\Delta t})}{C_\eta\,L^2}. \tag{19}$$

When $\varepsilon > \varepsilon_{\text{vac}}$, the bound provides no guarantee. This occurs when (i) $I(H^*; E_{t+\Delta t}) \approx 0$ (no precursors), or (ii) ε is large (poor pretraining), or (iii) L is large (irregular event boundary). Importantly, a vacuous bound does not imply that $I(H_t; E_{t+\Delta t}) = 0$; the encoder may retain information through paths not captured by our analysis (e.g. the encoder directly encodes precursor patterns without going through the predictor).

A.5 Why Predictor Weights Do Not Transfer

Our empirical finding (section 6.1) that predictor *architecture* matters but predictor *pretrained weights* do not is explained by a codomain mismatch.

During pretraining, $g_\phi : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^d$ maps encoder states to predicted *representations*; its codomain is the full d-dimensional representation space. During finetuning, the predictor (with the same architecture but different final layer) maps to *event logits*: $g'_\phi : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^K$, where K is the number of horizon intervals and $K \ll d$.

Formally, pretraining minimises $\mathbb{E}[\|g_\phi(H_t, \Delta t) - H^*\|_1]$, while finetuning minimises $\mathbb{E}[\ell_{\text{BCE}}(\sigma(g'_\phi(H_t, \Delta t)), E)]$. The pretraining objective rewards the predictor for reconstructing *all* d components of H^*, most of which encode event-irrelevant dynamics (channel-level forecasting, trend, seasonality). The finetuning objective rewards the predictor for extracting only the $\leq K$ bits relevant to event prediction. The optimal weight matrices for these two objectives need not be correlated, and indeed our experiments confirm they are not: at 100% labels, pred-FT with pretrained predictor weights achieves AUPRC within ± 0.003 of pred-FT with randomly initialised predictor weights.

This is analogous to the observation in vision that pretrained *classifier heads* do not transfer across tasks while pretrained *backbones* do: the backbone compresses the input into a general-purpose representation, while the head specialises to a specific output space.

A.6 Connection to Empirical Results

C-MAPSS (turbofan degradation). Degradation in turbofan engines develops over hundreds of cycles, with sensor readings progressively deviating from healthy baselines. The target encoder, seeing the future interval bidirectionally, captures degradation state with high fidelity: $I(H^*; E_{t+\Delta t})$ is large (the future interval is highly informative about whether failure occurs in that interval). Pretraining achieves low prediction error ε because the dynamics are smooth and near-deterministic. Proposition 1 predicts strong event-information retention, consistent with h-AUROC $= 0.74$ on FD002 and AUPRC $= 0.926$ on FD001.

CHB-MIT (seizure prediction). Seizure onset in scalp EEG has minimal electrographic precursors in the pre-ictal period, particularly within a 16-second context. The target encoder's representation H^* carries little event information: $I(H^*; E_{t+\Delta t}) \approx 0$. Proposition 2, condition (i), fails regardless of pretraining quality. This is consistent with the empirical result of h-AUROC $= 0.497$ (chance level).

FD002 (multi-operating-condition). FD002 adds operating-condition variability to FD001. This increases ε (harder prediction task) without proportionally increasing $I(H^*; E_{t+\Delta t})$ (event information is the same; only noise increases). The bound predicts that FD002 should be harder than FD001 for the same architecture, consistent with the observed drop from 0.926 to 0.908 AUPRC.

Label efficiency. At 5% labels, pred-FT achieves F1 $= 0.261$ vs. scratch $= 0.035$. The bound explains this: the encoder's $I(H_t; E_{t+\Delta t})$ is established during pretraining and does not depend on labels. Reducing labels degrades the finetuning optimisation (finding the right head weights) but cannot reduce the information available in the frozen encoder. Training from scratch must simultaneously learn both the representation and the event mapping, which is statistically harder with few labels.

A.7 Per-Horizon Generalisation

Proposition 1 is stated for a fixed horizon Δt. We now make the horizon dependence explicit. All quantities on the right-hand side of the bound can vary with Δt: the event posterior $\eta_{\Delta t}(h) = P(E_{t+\Delta t} = 1 \mid H^*_{\Delta t} = h)$, the Lipschitz constant $L(\Delta t)$, the prediction error $\varepsilon(\Delta t) = \mathbb{E}[\|H^*_{\Delta t} - \hat{H}_{\Delta t}\|_2^2]$, and the posterior bounds $\underline{\eta}_{\Delta t}, \overline{\eta}_{\Delta t}$ from **(A5)**.

Proposition 3 (Per-horizon information retention)**.** *Under the assumptions of proposition 1 applied at each horizon $\Delta t > 0$ separately, with horizon-dependent quantities $L(\Delta t)$, $\varepsilon(\Delta t)$, and $C_\eta(\Delta t) = (2\underline{\eta}_{\Delta t}(1 - \overline{\eta}_{\Delta t}))^{-1}$:*

$$I\big(H_t;\, E_{t+\Delta t}\big) \;\geq\; I\big(H^*_{\Delta t};\, E_{t+\Delta t}\big) \;-\; C_\eta(\Delta t)\, L(\Delta t)^2\, \varepsilon(\Delta t). \tag{20}$$

Proof. Apply the proof of proposition 1 verbatim at each fixed Δt, with all quantities carrying a subscript Δt. □

Predicted shape of the per-horizon AUROC curve. Equation (20) predicts how the encoder's event information varies with horizon. Three effects combine:

- $I(H^*_{\Delta t}; E_{t+\Delta t})$ is typically non-monotone: very short horizons may not yet encompass precursors; very long horizons dilute the event signal with unrelated dynamics.
- $\varepsilon(\Delta t)$ is monotonically increasing (representations further into the future are harder to predict from H_t).
- $L(\Delta t)$ can increase with Δt if the event boundary sharpens as the event approaches.

The net effect is that $I(H_t; E_{t+\Delta t})$ peaks at intermediate Δt and decays at large Δt. While MI and AUROC are not monotonically related in general, this qualitative shape is compatible with the per-horizon AUROC curves observed in our experiments: AUROC tends to be highest at $\Delta t \in [10, 50]$ cycles for C-MAPSS and at $\Delta t \in [1, 20]$ steps for anomaly datasets, then declines. The decay is more pronounced for the anomaly datasets because $\varepsilon(\Delta t)$ grows faster when the dynamics are irregular.

A.8 Relationship to the Information Bottleneck

The Information Bottleneck (IB) framework [Tishby et al., 2000] seeks a representation T of input X that maximises $I(T; Y)$ (relevance) while minimising $I(T; X)$ (compression):

$$\min_{P(T|X)} \; I(T; X) - \beta\, I(T; Y). \tag{21}$$

JEPA pretraining differs from IB in three ways:

1. **No explicit compression.** JEPA does not penalise $I(H_t; X_{\leq t})$. The encoder is free to retain all information about the past. Implicit compression arises only from the architecture bottleneck ($d = 256$).
2. **Predictive, not discriminative.** The IB target Y is a label; the JEPA target is a future representation H^*. JEPA maximises $I(H_t; H^*)$ (through the predictor), which is an upper bound on $I(H_t; Y)$ for any Y that is a function of the future interval (by DPI). This makes JEPA a *looser* but *more general* objective: it retains information about all downstream tasks, not just one.
3. **Asymmetric architecture.** The IB is typically symmetric in X and Y. JEPA imposes causal structure: the encoder sees only the past, the target encoder sees only the future. This causal asymmetry is essential for event prediction, where we cannot condition on the future at test time.

The connection to our result is direct. Let $Y = E_{t+\Delta t}$. By proposition 1:

$$I(H_t; Y) \;\geq\; I(H^*; Y) \;-\; C_\eta\, L^2\, \varepsilon. \tag{22}$$

As the JEPA pretraining loss drives $\varepsilon \to 0$, the encoder's mutual information with Y approaches $I(H^*; Y)$, the maximum achievable by the target representation. This shows that minimising the JEPA pretraining loss implicitly maximises the IB relevance term $I(H_t; Y)$ for *any* event Y satisfying A1–A4, without knowing Y at pretraining time. The price paid relative to IB is that JEPA also retains event-irrelevant information (it does not compress), which manifests as higher representation dimensionality but not as reduced downstream performance.

B Horizon Interval Design

FAM uses $K = 150$ densely-sampled horizon intervals $\Delta t \in \{1, 2, \ldots, 150\}$ during both pretraining and downstream evaluation. This design choice ensures the probability surface $p(t, \Delta t)$ captures fine-grained temporal structure. Foundation model baselines (Chr-2, MOMENT, TimesFM, Moirai) use $K = 7$ horizons $[1, 5, 10, 20, 50, 100, 150]$ for C-MAPSS and $K = 8$ for anomaly datasets to avoid sequence-length bottlenecks in the frozen backbone. The h-AUROC evaluation uses whichever horizons are available, skipping degenerate horizons (event prevalence < 0.001 or > 0.999). A full horizon-count sweep is deferred to a future version.

C Architectural Decision Log

Decision	Value	Source	Status
Attention mask	Causal	ablation study	Fixed, not compared
Horizon sampling	LogU$[1, 150]$	initial sweep	Default
Target interval	Cumulative $(t, t{+}\Delta t]$	ARCHITECTURE.md	Default
Output param.	Discrete hazard $\rightarrow$ CDF	section 3.4	Default
EMA momentum	$\tau = 0.99$	initial sweep	Default
Encoder depth	$L = 2$	initial sweep	Default (no sweep)
d_{model}	256	initial sweep	Default (no sweep)
Loss	L1 on L2-normalised	initial sweep	Default
Target encoder	EMA or SIGReg	table 9	SIGReg-pred wins

D Full Chronos-2 Comparison

Table 6 compares FAM and Chronos-2 as frozen feature extractors under two head configurations: (a) matched 198K dt-conditioned MLP (the primary comparison), and (b) linear probes for reference (unmatched capacity). Both models use the same labels, loss, and evaluation protocol; only the frozen encoder differs. FAM and Chronos-2 do not use Chronos-2's native forecasting capability in any evaluation below.

Table 6: **FAM vs. Chronos-2: h-AUROC under matched head (198K MLP, v30 Phase 1).** Sparse horizons K=8. SMD omitted (Chronos-2 feature extraction is infeasible at this sequence length). MSL excluded from main table (FAM h-AUROC 0.35, below chance).

Dataset	Model	h-AUROC (198K MLP)	h-AUROC (probe)	Probe params/horizon
FD001	FAM	$\mathbf{0.707 \pm 0.018}$	$\mathbf{0.742 \pm 0.012}$	257
FD001	Chronos-2	0.659 ± 0.000	0.622 ± 0.004	769
FD003	FAM	$\mathbf{0.788 \pm 0.026}$	$\mathbf{0.812 \pm 0.006}$	257
FD003	Chronos-2	0.760 ± 0.003	0.738 ± 0.002	769
MBA	FAM	$\mathbf{0.721 \pm 0.021}$	0.588 ± 0.012	257
MBA	Chronos-2	0.451 ± 0.017	$\mathbf{0.659 \pm 0.006}$	769
BATADAL	FAM	$\mathbf{0.566 \pm 0.009}$	0.521 ± 0.038	257
BATADAL	Chronos-2	0.534 ± 0.032	0.503 ± 0.040	769
FAM wins (198K MLP)	4 / 4 datasets			
FAM wins (probe)	3 / 4 datasets (MBA: Chr-2 probe wins; FAM pred-FT wins)			

Probe comparison (unmatched capacity): FAM probe has 257 trainable params/horizon; Chr-2 probe has 769. The probe comparison confirms FAM encoder quality but is not the primary claim. The 198K MLP comparison uses identical capacity for both; FAM wins 4/4. The MBA anomaly: Chr-2 probe (769 p/h) beats FAM probe (257 p/h), but FAM pred-FT (198K total) beats Chr-2 MLP (198K total). This indicates Chronos-2's 768-dim embeddings encode MBA arrhythmia information that linear probes can exploit with more params, but the 198K MLP cannot effectively leverage the high-dim space. FAM's lower-dim encoder (256-d) is more efficiently exploited by the matched MLP head.

E MOMENT-1-large Comparison

MOMENT-1-large [Goswami et al., 2024] (341.2M parameters) is a general time-series foundation model pretrained on MIMIC-III and diverse public datasets. To isolate encoder quality from head

capacity, we apply the same 198K-param dt-conditioned MLP head as chr2-mlp, with identical training protocol (100% labels, 3 seeds). Context windows are processed per-channel (univariate, as MOMENT requires), producing 1024-d embeddings that are mean-pooled across channels for multivariate inputs. Evaluation uses per-horizon mean AUROC, skipping degenerate horizons (prevalence < 0.001 or > 0.999). MOMENT requires per-channel (univariate) processing; at inference time each forward pass processes $B \times C$ sequences simultaneously. For SMD (C=38, 28 entities), this requires $\approx$10.6 h extraction, which we deem infeasible; all other 10 datasets in our benchmark are evaluated.

Table 7: **MOMENT-1-large vs. FAM and Chronos-2 (198K MLP head, 3 seeds, h-AUROC; SMD excluded: $\approx$10.6 h B$\times$C extraction).** Results shown for 5 datasets (FD001-FD003, BATADAL, MBA); SMAP, PSM, GECCO, SKAB, ETTm1 extension in progress. FAM wins on 3 of 5 shown datasets. MOMENT wins MBA (MIMIC-III pretraining overlaps cardiac ECG) and FD002 (multi-condition turbofan where large-corpus pretraining helps); Chronos-2 also wins FD002 for the same reason. Per-channel univariate processing discards cross-channel dependencies on high-dimensional industrial datasets (FD001, FD003, BATADAL), where domain-specific JEPA pretraining dominates.

Dataset	MOMENT-mlp	Chr2-mlp	FAM
FD001	0.559 ± 0.009	0.659 ± 0.002	$\mathbf{0.786 \pm 0.033}$
FD002	0.704 ± 0.003	$\mathbf{0.734 \pm 0.001}$	0.566 ± 0.011
FD003	0.473 ± 0.012	0.760 ± 0.003	$\mathbf{0.853 \pm 0.004}$
BATADAL	0.537 ± 0.066	0.534 ± 0.032	$\mathbf{0.607 \pm 0.033}$
MBA	$\mathbf{0.791 \pm 0.009}$	0.451 ± 0.017	0.739 ± 0.014

FD003 MOMENT h-AUROC 0.473 is below chance (0.5), indicating per-channel univariate embeddings anti-predict turbofan degradation horizons. All methods use the same 198K-param dt-MLP head; encoder quality is the only variable. MOMENT uses K=7 horizons $[1, 5, 10, 20, 50, 100, 150]$ for C-MAPSS and K=8 for BATADAL/MBA; FAM uses K=150 dense horizons. BATADAL MOMENT std=0.066 reflects high seed sensitivity consistent with BATADAL's single-entity anomaly pattern. Per-seed: FD001 (0.568, 0.561, 0.549), FD002 (0.700, 0.707, 0.704), FD003 (0.482, 0.459, 0.478), BATADAL (0.568, 0.462, 0.582), MBA (0.801, 0.780, 0.791). FD002 win for MOMENT and Chr-2: FD002 has 6 operating conditions; large-corpus pretraining captures cross-condition variation that per-dataset JEPA (trained on single-condition data) encodes less efficiently. MBA win for MOMENT is consistent with MIMIC-III pretraining overlap and low channel count (C=2). SMD excluded: $B \times C$ inference requires $\approx$10.6 h extraction (38 channels $\times$ 28 entities). SMAP/PSM/GECCO/SKAB/ETTm1 feasible; extension in progress.

F TimesFM and Moirai Comparisons

TimesFM-1.0-200M [Das et al., 2024] (203.6M parameters) and Moirai-1.1-R-base [Woo et al., 2024] (91.4M parameters) are large-scale time series foundation models pretrained for value forecasting. To isolate encoder quality from head capacity, we extract frozen representations and train an identical 198K-param dt-conditioned MLP head using the same labels, loss, and evaluation protocol as the main Chr-2 and MOMENT comparisons (3 seeds each). TimesFM representations are extracted via a forward hook on the final transformer block (mean-pooled over patches, producing 1280-d embeddings); Moirai representations are extracted via a hook on the encoder output (768-d embeddings). FAM uses dense K=150 horizons; baselines use $K \in \{7, 8\}$ horizons to avoid sequence-length bottlenecks.

G Additional Ablations

G.1 Does the Predictor Help at Inference?

With all weights frozen, probe_h ($\mathbf{h}_{\text{past}}$ only) achieves 0.299 F1w at 100% labels, while frozen_multi ($[\hat{\mathbf{h}}_1; \ldots; \hat{\mathbf{h}}_{16}]$) achieves only 0.148. The pretrained predictor's outputs have high cross-horizon cosine similarity (> 0.98), so the concatenation is nearly redundant. Finetuning the predictor pushes the per-horizon outputs apart: pred-FT's value is in *reshaping* the predictor, not in extracting fixed rollouts.

G.2 Pretrained vs. Random-Init Predictor

Encoder frozen; toggle only predictor initialisation: pretrained vs. random-init, then pred-FT identically (3 seeds each):

Table 8: **TimesFM-1.0-200M and Moirai-1.1-R-base vs. FAM (198K MLP head, 3 seeds, h-AUROC, lf=100%).** All models use identical downstream head, labels, and evaluation. FAM wins on physical-degradation datasets (FD001, FD003, ETTm1, SMAP) by large margins; TimesFM wins on short-burst anomaly datasets (GECCO, BATADAL, SKAB, MBA, PSM, FD002, SMD). SMD TimesFM margin is small ($+0.012$, std 0.019 vs FAM 0.004). Moirai evaluated on 9 feasible datasets (SMAP and SMD infeasible: $B \times C$ extraction ≥ 4.8M sequences). FAM wins 6 of 9; Moirai wins FD002 ($+0.098$), SKAB ($+0.116$), and GECCO ($+0.003$, effectively tied, overlapping CIs). Moirai falls below chance on BATADAL (0.360). "—": extraction infeasible (SMAP, SMD).

Dataset	TimesFM-mlp	Moirai-mlp	FAM
FD001	0.530 ± 0.003	0.606 ± 0.004	$\mathbf{0.786 \pm 0.033}$
FD002	0.602 ± 0.008	$\mathbf{0.664 \pm 0.004}$	0.566 ± 0.011
FD003	0.615 ± 0.014	0.700 ± 0.004	$\mathbf{0.853 \pm 0.004}$
SMAP	0.505 ± 0.028	—	$\mathbf{0.598 \pm 0.036}$
PSM	$\mathbf{0.570 \pm 0.007}$	0.533 ± 0.006	0.562 ± 0.013
MBA	$\mathbf{0.759 \pm 0.006}$	0.571 ± 0.017	0.739 ± 0.014
GECCO	$\mathbf{0.925 \pm 0.006}$	0.822 ± 0.008	0.819 ± 0.064
BATADAL	$\mathbf{0.653 \pm 0.005}$	0.360 ± 0.010	0.607 ± 0.033
SKAB	0.744 ± 0.010	$\mathbf{0.823 \pm 0.001}$	0.707 ± 0.017
ETTm1	0.589 ± 0.008	0.597 ± 0.003	$\mathbf{0.869 \pm 0.002}$
SMD	$\mathbf{0.665 \pm 0.019}$	—	0.654 ± 0.004

Per-seed TimesFM (lf=100%): FD001 (0.5279, 0.5291, 0.5342), FD002 (0.5921, 0.6013, 0.6111), FD003 (0.5973, 0.6165, 0.6324), SMAP (0.4834, 0.4865, 0.5441), PSM (0.5637, 0.5659, 0.5791), MBA (0.7535, 0.7558, 0.7682), GECCO (0.9165, 0.9268, 0.9320), BATADAL (0.6455, 0.6554, 0.6568), SKAB (0.7338, 0.7410, 0.7586), ETTm1 (0.5797, 0.5881, 0.5992), SMD (0.6545, 0.6503, 0.6916). Per-seed Moirai (lf=100%): FD001 (0.6021, 0.6054, 0.6105), FD002 (0.6615, 0.6617, 0.6696), FD003 (0.6962, 0.6975, 0.7060), MBA (0.5468, 0.5797, 0.5858), GECCO (0.8110, 0.8258, 0.8298), BATADAL (0.3477, 0.3591, 0.3725), SKAB (0.8225, 0.8226, 0.8252), ETTm1 (0.5931, 0.5973, 0.5997), PSM (0.5280, 0.5303, 0.5412). TimesFM context: 512 time steps per channel (univariate), patch size 32; Moirai: 512 time steps, patch size 32, max patch size 128. BATADAL: Moirai's univariate patching discards multivariate cross-sensor correlations critical for this dataset (0.360, below chance). FD002 TimesFM win: 6 operating conditions; large-corpus pretraining captures cross-condition variation. GECCO TimesFM win: univariate time-of-day patterns well-covered by general pretraining corpus. FAM uses K=150 dense horizons; baselines use K=7 (C-MAPSS) or K=8 (anomaly datasets).

- **FD001**: pretrained 0.9257 ± 0.0008 vs. reset 0.9235 ± 0.0027, $p = 0.38$.
- **SMAP**: pretrained 0.3874 ± 0.0205 vs. reset 0.3950 ± 0.0286, $p = 0.34$.
- **MBA**: pretrained 0.9465 ± 0.0004 vs. reset 0.9435 ± 0.0016, $p = 0.089$.

Pretrained predictor weights carry at most $+0.003$ AUPRC. The value of pred-FT is in the pretrained *encoder* and the finetuning recipe, not the predictor's initial weights.

G.3 EMA vs. SIGReg Pretraining

SIGReg [Balestriero and LeCun, 2025] replaces the EMA target encoder with spectral regularisation. On FD001 (5 seeds, pred-FT downstream):

Table 9: **Pretraining objective ablation (FD001, 5 seeds).** SIGReg-pred wins on RMSE at both label budgets.

	100% labels		**5% labels**	
Pretraining	F1w ↑	RMSE ↓	F1w ↑	RMSE ↓
EMA (default)	0.391 ± 0.085	16.90 ± 1.71	0.261 ± 0.165	24.33 ± 6.83
SIGReg-enc	0.401 ± 0.070	14.08 ± 0.98	0.252 ± 0.156	17.52 ± 1.34
SIGReg-pred	$\mathbf{0.451 \pm 0.064}$	$\mathbf{13.71 \pm 0.34}$	0.243 ± 0.165	$\mathbf{17.30 \pm 3.55}$

Table 10: **Sub-5% label efficiency on FD001 and FD003 (FAM-predft, 3 seeds each).** Retention is relative to full-label h-AUROC. FD001 retains 92% with just 2 engines (2% of 85). FD003 degrades faster due to multi-fault variability.

Dataset	100%	10%	5%	2%	1%
FD001 (85 eng.)	0.786 ± 0.033 100%	0.772 ± 0.059 98%	0.730 ± 0.018 93%	$\mathbf{0.724 \pm 0.013}$ **92%**	0.670 ± 0.110 85%
FD003 (100 eng.)	0.853 ± 0.004 100%	0.830 ± 0.018 97%	0.709 ± 0.131 83%	0.635 ± 0.065 74%	0.513 ± 0.220 60%†

FD001 at 1%: single-engine sampling introduces high variance (s42=0.544, s123=0.731, s456=0.737). †FD003 at 1% (1 engine): mean 0.513±0.220; per-seed s42=0.292 (near-chance - unlucky engine), s123=0.731, s456=0.515. With std=0.220, the 1-engine result is unreliable and is included for completeness only. FD003's multi-fault modes (4 fault types) require multiple engines for reliable representation.

Table 11: **Legacy label-efficiency curve (F1w, legacy backbone).** FD001 at 5/10/20%: N=10 paired seeds; others N=5. Bold: $p < 0.05$; italic: $p < 0.15$. Under the canonical architecture the crossover does not survive (see section 6.2).

Subset	Budget	N	Pred-FT F1w	E2E F1w	paired p	collapses p/e
FD001	100%	5	0.391 ± 0.085	0.408 ± 0.119	0.84	0/0
	50%	5	0.391 ± 0.083	0.356 ± 0.074	0.27	0/0
	20%	10	0.276 ± 0.184	0.312 ± 0.102	0.57	2/0
	10%	10	$\mathbf{0.291 \pm 0.120}$	0.133 ± 0.164	**0.023**	0/4
	5%	10	*0.229 ± 0.141*	0.124 ± 0.201	*0.114*	1/7
FD002	100%	5	0.315 ± 0.097	$\mathbf{0.426 \pm 0.046}$	**0.038** (E2E wins)	0/0
	10%	5	0.258 ± 0.059	0.259 ± 0.048	0.98	0/0
	5%	5	*0.289 ± 0.120*	0.216 ± 0.133	*0.070*	0/0
FD003 (200 ep)	**100%**	5	$\mathbf{0.270 \pm 0.059}$	0.123 ± 0.166	**0.043**	0/2
	10%	5	0.043 ± 0.059	0.096 ± 0.079	0.22	3/1
	5%	5	0.062 ± 0.082	0.009 ± 0.021	0.25	1/4

G.4 Sub-5% Label Efficiency (FAM-predft, h-AUROC)

G.5 Full Label-Efficiency Curve (Legacy Backbone)

H Extended Results

Per-seed raw results for each table are available in the code repository as JSON files with per-horizon AUROC, precision, recall breakdowns at each horizon interval.

I PA-F1 Inflation

PA-F1 (Point-Adjusted F1) credits an entire anomaly segment as correctly predicted if any single timestep within it exceeds the threshold [Xu et al., 2022]. This protocol inflates reported F1 dramatically when anomaly segments are long and prevalence is non-trivial: the TSAD-Eval study [Schmidl et al., 2022] shows that a random detector can exceed $F_1 = 0.9$ PA on some datasets. We concretely demonstrate this by contrasting PA-F1 and non-PA F1 for FAM+Mahalanobis on our benchmarks: SMAP 0.793 PA vs. 0.038 non-PA, SMD 0.252 PA vs. 0.144 non-PA, PSM 0.813 PA vs. 0.085 non-PA. The gap in PA-F1 between FAM and MTS-JEPA remains large even under a random-baseline sanity check (a random-init encoder + Mahalanobis baseline achieves 0.604 ± 0.007 PA-F1 on SMAP; FAM is 0.79), so the PA-F1 margin is not purely inflation - but any single PA-F1 number should be read in conjunction with the corresponding non-PA number.

J Preprocessing Details

Dataset	Channels	Preprocessing	Context
C-MAPSS FD001-003	14 sensors (after drop)	per-subset min-max	cycle-as-patch
SMAP	25 (after drop)	z-score on train	100 steps
PSM	25 (after drop-constant)	pre-normalised	100 steps
SMD	38 per-machine	pre-normalised	100 steps
MBA ECG	2 leads	z-score on train	100 steps
GECCO water	9 quality sensors	z-score on train	512 steps
BATADAL SCADA	43 tank/pump/pressure	z-score on train	512 steps
SKAB hydraulic	8 pressure/flow sensors	z-score on train	100 steps
ETTm1 power	7 load/temperature	z-score on train	512 steps

The normalisation, channel-drop thresholds, and context lengths follow the conventions in the cited SSL and anomaly-detection benchmarks; specific numeric thresholds are in `fam-jepa/data/*.py`. For C-MAPSS we use the SELECTED_SENSORS subset from our preprocessing pipeline (14 sensors after removing near-constant channels). RUL cap is 125 cycles, following STAR.